

SHEN, Dingtao

Tel: +852 60868542 | Email: dtshen@outlook.com | Homepage: <https://dingtao-shen.github.io/>

Education

The Hong Kong University of Science and Technology

Ph.D. *The Division of Emerging Interdisciplinary Areas, Academy of Interdisciplinary Studies* 2023.8 – 2027.7(Expected)

- Focus: computational math & physics; numerical algorithms/analysis; high-fidelity phonon BTE solutions; ML/AI for scientific computing
- Redbird Academic Scholarship (2024-2025 / 2025-2026)

University of Wisconsin-Madison

Master *Department of Mathematics (Applied & Computational Mathematics)* 2021.8 – 2023.5

- Core Coursework: Applied Math / Probability / Stochastic Computation Methods / Nonlinear Optimization, etc.

Zhejiang University

Bachelor *School of Mathematical Sciences (Information & Computing Science)* 2017.9 – 2021.6

- ZJU Outstanding Academic Performance Scholarship (2017-2018)

Publication & Research

- **Shen, Dingtao**, Jia Liu, and Wei Su. "A Synthetic Iterative Scheme for Non-Gray Phonon Boltzmann Transport Equation with Dual Relaxation Times." *arXiv preprint arXiv:2608.28101* (2026). (Submitted to *Journal of Computational Physics*)
 - Developed a macroscopic synthetic iteration scheme to accelerate the conventional source iteration for high-dimensional, mode-dependent phonon BTE with dual relaxation times
 - Performed algorithm design, convergence/stability and asymptotic-preserving analysis, and numerical experiments
- **Shen, Dingtao**, and Wei Su. "Accurate-geometry-embodied finite element method for phonon Boltzmann transport equation." *Computer Physics Communications* 313 (2025): 109623.
 - The AGEFEM for high-dimensional phonon BTE substantially reduces discretization error in complex geometries
 - Talks & Presentations: Minisymposium @ EASIAM 2024; HKUST SIAM Student Chapter 2025
- A General High-fidelity FEM Solver for Phonon Boltzmann Transport Equation
 - General high-performance solver for phonon BTE across single/dual-relaxation-time models, 2D/3D, and complex geometries
 - AI-assisted performance/engineering optimization & project management: <https://github.com/dingtao-shen/FEMpBTE>
- (on-going) StKine-Bench: An Agent Benchmark for Stiff Kinetic Transport Solver Development
 - AI for Scientific Computing: AI-agent benchmark framework for kinetic transport solver development
 - Agent evaluations of dual-relaxation gray phonon BTE acceleration produced a physics-preconditioned GMRES solver
- Option Sensitivity Estimation and Differential Machine Learning under Discontinuous Payoffs
 - Risk-neutral GBM framework for option pricing/Delta estimation; analyzed bias-variance behavior under discontinuous payoffs
 - Trained a joint price/sensitivity model; used independent parameter tests and fixed-budget simulations to assess how label quality affects pricing and risk-approximation error

Teaching & Work Experience

The Hong Kong University of Science and Technology · EMIA - Teaching Assistant

- Graduate Course: EMIA6500A Special Topics: Discontinuous Galerkin Methods (2024 Spring)
- Undergraduate Course: EMIA2020 Cross-disciplinary Design Thinking (2024 Fall / 2026 Spring)

Zhejiang University · School of Mathematical Sciences - Research Assistant 2021.7 – 2022.6

- Project: Error analysis for Local DG with implicit-explicit (IMEX) multistep discretization of the nonlinear Schrödinger equation

Skills

- Programming & Software: C++, Python, MATLAB, SQL; Git, Linux 等
- AI Skills: AI-driven programming (Vibe Coding); AI-assisted development tools (Claude Code/Codex)
- Language: Chinese, English